



Studybeesofficial

Where Minds Pollinate

Computer Science
Class 10



FBISE

Chapter 2
Notes

Follow Our Socials



studybeesofficial



studybeesofficialpak



studybeesofficialpak

Computer Science 10th
Chapter no 2
SLO Based Questions(Q1-9)

Qno1: What is Computational Thinking(CT) ?

Ans: Computational Thinking (CT) is a way of solving problems by using ideas from computer science. It helps you break down big problems into smaller parts and come up with step-by-step solutions that a computer or a person can follow. It involves using mental tools and strategies to think more clearly, logically, and creatively.

Qno2: Explain what are Computing Problems?

A computing problem in computer science is one that is solved step-by-step through computation. Any kind of calculation, including arithmetical and logical ones, can be included.

Following are some types/domains of Computing Problems

1. Decision Problems
2. Searching Problems
3. Counting Problems

Qno3: Explain some domains of Computing Problems.
OR

Explain the types of Counting Problems

Ans: Some of the computing problem domains are as follows:

Decision Problems

In decision problems normally 'yes-or-no' response is found in a choice.

For example, given a number n, "is n even?"

Search Problems

In search problems, the solution is made up by having a set of values that satisfy a certain criteria.

For example: we might wish to calculate a route on a map from one site to another (e.g. from Islamabad to Lahore)

Counting Problems

In counting problems, the number of possible solutions to a search problem is the answer of that counting problem. For the above-mentioned search problem "we might wish to calculate a route on a map from one site to another." The answer of counting problem would be two, supposing we have two roads from Islamabad to Lahore e.g. G.T. Road and Motorway.

Qno4: Briefly explain some basic counting problems.

Ans: Basic Counting Principles

Addition

The addition principle of counting is a basic idea for counting the total number of possibilities that can occur when dealing with many events or options that are mutually exclusive. The events that are mutually exclusive are those that cannot occur simultaneously.

Total number of possibilities = Number of possibilities for event A + Number of possibilities for event B

For example: a student wants to buy a new computer. He has to decide among 3 desktop computers and 4 laptop computers.

Total number of computer options = Total number of desktop computer options + Total number of Laptop options

Total number of computer options = $3 + 4 = 7$

Multiplication

The Multiplication Principle is used when we are dealing with many events, and we have to select one option from each event.

For example: Bilal has 3 Apples, 3 Bananas and 3 bunch of Cherries. In how many ways, he can make fruit basket by putting fruit of one kind.

Let Apples be represented by **A**, Bananas be represented by **B**, Cherries be represented by **C**

Options available for Apples (A) = 3

Options available for Bananas (B) = 3

Options available for Bunch of Cherries (C) = 3

Total ways to make fruit baskets are = $3 * 3 * 3$

Possible ways are =

(A1,B1,C1), (A1,B1,C2), (A1,B1,C3), (A1,B2,C1), (A1,B2,C2), (A1,B2,C3), (A1,B3,C1), (A1,B3,C2), (A1,B3,C3),
(A2,B1,C1), (A2,B1,C2), (A2,B1,C3), (A2,B2,C1), (A2,B2,C2), (A2,B2,C3), (A2,B3,C1), (A2,B3,C2), (A2,B3,C3),
(A3,B1,C1), (A3,B1,C2), (A3,B1,C3), (A3,B2,C1), (A3,B2,C2), (A3,B2,C3), (A3,B3,C1), (A3,B3,C2), (A3,B3,C3),

Permutation

A **permutation** is a way to arrange a specific number of items in a **certain order**. It applies to problems where putting objects in order matters, such as arranging letters into words, digits into numbers, or choosing and positioning items sequentially.

Combination

A **combination** is a method of choosing a few items at random from a collection **without regard to the order**. It applies to situations where you select items and their final sequence or arrangement is

irrelevant.

Pigeonhole Principle

The principle is stated as:

There are (n) boxes (n) is a positive integer) and (n + 1) objects are to be placed into these (n) boxes, then at least one box contains two or more objects.

Example

Suppose you have 13 apples and 12 baskets. According to the Pigeonhole Principle, at least one basket will contain more than one apple.

Inclusion and Exclusion

If there are some ways in (n) and (m) to do the task, then according to the inclusion-exclusion principle, we must sum the number of ways to complete a task one way and the number of ways to complete it another, and then subtract the number of ways that are shared by both sets of ways in order to count only unique ways of completing the work.

$$N(A \cup B) = N(A) + N(B) - N(A \cap B)$$

Qno5: How would you explain “Algorithm”?

An algorithm is a well-defined, step-by-step procedure or set of rules for solving a specific problem or performing a task. They can be expressed in a variety of ways, including computer languages, flowcharts, pseudo code, and natural language.

Qno6: Write Properties of algorithm

Properties of Algorithms

Input: An algorithm can have one or more externally supplied values as inputs.

Output: The outcome of the calculations made by an algorithm is at least one output.

Definiteness: Every algorithmic step needs to be clearly and accurately stated.

Finiteness: There must always be a finite number of steps in an algorithm before it ends.

Effectiveness: An algorithm's operations must be simple enough for a human to execute them with paper and pencil in a reasonable amount of time.

Generality: An algorithm should be able to handle a group of problems rather than simply one particular one. It needs to offer a broad solution for several related problems.

Qno7: Explain Role of Algorithms in Computational Problem-Solving.

Ans: Role of Algorithms in Computational Problem-Solving

- Algorithms are designed to be efficient in terms of time and space.

- Algorithms automate repetitive and tedious tasks
- Algorithms help in analyzing and breaking down complex problems into manageable steps.
- Once an algorithm is designed and tested, it can be reused for similar problems.
- Algorithms provide precise and accurate solutions
- Algorithms are considered foundation of programming and software development.

Qno8: Write some applications of algorithms.

The algorithms can be applied to solve various problems such as

Sorting and Searching: For organizing and retrieving data efficiently, algorithms like quicksort, mergesort, and binary search are used.

Graph Algorithms: For finding the shortest paths in navigation applications, Dijkstra algorithm could be used.

Cryptography: TO provide privacy and security in online transactions, such as algorithms be used to encrypt and decrypt data.

Machine Learning: To learn from data and make predictions, machine learning algorithms like decision trees, neural networks can be used.

Image Processing: To enhance image quality or compress the image to reduce size or even to analyze images, Image processing algorithms could provide help. These have very critical role in medical imaging, computer vision, and multimedia applications.

Qno9: How can be the Logical reasoning fit into computational thinking?

Ans: Logical reasoning fits into computational thinking in the following ways:

Problem Decomposition: Logical reasoning is needed to guarantee that each component is treated systematically when breaking large problems down into smaller, more manageable portions.

Pattern Recognition: Logical sequences and linkages are frequently recognized in order to identify patterns in data or processes.

Abstraction: Logical thinking is needed to discern between pertinent and extraneous information in order to simplify complicated problems by concentrating on the most important features.

Algorithm Design: To guarantee the accuracy and effectiveness of the process, logical thinking is a crucial component of creating step-by-step solutions, or algorithms, to issues.

Qno10: Define Boolean Logic and explain the difference between the AND operator and the OR operator.

In Boolean, two statements or expressions are evaluated in a way that can only have one of two potential values as a result—either true or false. Boolean logic uses logical operator like AND, OR, NOT, etc.

Truth Table of Boolean operators

A	B	A AND B	A OR B	NOT A
False	False	False	False	True
False	True	False	True	True
True	False	False	True	False
True	True	True	True	False